POCKETPT: USER MANUAL AND INSTALLATION GUIDE

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INTRODUCTION

1.1 About PocketPT

PocketPT is a comprehensive mobile and web-based physical therapy application designed to guide users through personalized rehabilitation programs. The app combines assessment tools, exercise planning, real-time recording with AI-powered pain detection, and progress tracking to deliver a complete rehabilitation experience.

1.2 Key Features

- ✓ **Personalized Assessment:** Multi-step evaluation to determine your rehabilitation needs
- ✓ **Custom Exercise Plans:** CSV-backed exercise library with personalized weekly programs
- ✓ **AI-Powered Recording:** Real-time facial pain recognition and pose detection during exercises
- ✓ **Treatment Recommendations:** Severity-based treatment plans including modalities and interventions
- ✓ **Progress Tracking:** Comprehensive reports with charts, calendars, and exportable PDFs
- ✓ **Guest Mode:** Try the app without creating an account (web)

1.3 System Requirements

Mobile (Android)

- ✓ Android 7.0 (Nougat) or higher
- ✓ Minimum 2GB RAM
- ✓ Camera
- ✓ 750MB free storage space

Web Browser

- ✓ Modern browsers: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- ✓ Stable internet connection recommended
- ✓ Camera access for recording features

GETTING STARTED

2.1 Creating Your Account

1. Open PocketPT on your device.

2. Tap "Create Account" on the login screen.

3. Fill in your information:

   - ✓ Full name
   - ✓ Email address
   - ✓ Secure password (minimum 8 characters)

4. Tap "Create Account".

5. Verify your email:

   - ✓ Check your inbox for a verification email
   - ✓ The app will automatically poll for verification
   - ✓ Once verified, you'll be signed in automatically

2.2 Guest Mode

If you want to try PocketPT without creating an account:

1. On the login screen, tap "Continue as Guest"
2. Your data will be stored locally on your device
3. Important: Guest mode data is not synced to the cloud and will be lost if you clear browser data

2.3 First-Time Setup

After logging in for the first time:

1. Grant Permissions:
   - Camera access (for recording and pose detection).
   - Notifications (optional, for reminders).
2. Complete Initial Assessment (optional):
   - The dashboard will prompt you to start your assessment.
   - You can skip and do this later from the dashboard.

USER MANUAL

3.1 Navigation Overview

PocketPT uses a bottom navigation bar with five main sections:

Icon	Section	Purpose
	Dashboard	View your plans, progress, and notifications
	Plan Manager	Manage exercises and treatments
	Record	Start exercise recording sessions
	Reports	View analytics and export PDFs

3.2 Dashboard

The Dashboard serves as your central home base within the application, providing a comprehensive overview of your rehabilitation journey and quick access to essential features. Upon opening the app, this screen automatically loads your latest data, ensuring you always see your current status.

Hero Card

The Hero Card is prominently displayed at the top of the Dashboard. It clearly shows your assessment completion status, indicating whether you have an active plan or need to begin the process. This card also provides a quick action button, allowing you to easily start a new assessment or continue an existing one if it was previously incomplete.

Plan Overview

Your current rehabilitation plan is presented in a clear Plan Overview section. This area displays your personalized plan with a structured week-by-week breakdown, helping you understand your progression over time. Each exercise is shown on individual cards that detail the exercise name along with the prescribed sets and repetitions. Directly alongside your exercises, you will find your personalized treatment recommendations, which provide guidance on complementary therapies like ice or heat application.

Progress Stats

To help you track your recovery, the Dashboard features a section with key Progress Stats. These metrics include your most recently reported pain level, your overall exercise completion rate, and a count of your consecutive days active streak. These statistics offer valuable insights into your consistency and overall progress.

Notification

A Notification Center consolidates important updates and reminders into actionable cards. You may receive alerts prompting you to complete your daily pain assessment, notifications about new plan

regeneration alerts, or other system messages that require your attention, ensuring you never miss a critical update.

Key Actions:

- Start Assessment: Begin your rehabilitation evaluation
- Start Recording: Jump directly to exercise recording
- View Plan Details: Expand any exercise or treatment card for full information

3.3 Assessment

The assessment is the foundation of your personalized rehabilitation journey within PocketPT. This step-by-step process gathers essential information about your condition, goals, and physical state to automatically generate a safe and effective plan tailored just for you. You will progress through a series of intuitive screens, each designed to capture a specific aspect of your health, culminating in a customized exercise and treatment program.

Preliminary Information

Your journey begins with an overview of the assessment process. This screen explains what kind of data will be collected—such as your goals, pain levels, and movement limitations—and how this information will be used to build your rehabilitation plan. It ensures you are fully informed and comfortable before you start, setting clear expectations for the steps ahead.

Rehabilitation Goal

Here, you define the primary objective of your therapy. You will choose from key rehabilitation goals, including pain reduction, mobility improvement, strength building, or returning to a specific sport or activity. This selection is crucial as it directs the entire focus of your generated plan, ensuring the recommended exercises and treatments align with your personal recovery ambitions.

Focus Area

Using interactive anatomical diagrams, you pinpoint the exact muscle group or body area requiring rehabilitation. The interface allows you to tap on specific regions of the body for selection. If your condition is one-sided, you will have the option to specify whether it affects the left or right side, guaranteeing that your plan targets the correct location with precision. Muscles involved include:

- ✓ Deltoids
- ✓ Biceps
- ✓ Triceps
- ✓ Abdominals
- ✓ Obliques
- ✓ Lower Back
- ✓ Multifidus
- ✓ Quadriceps
- ✓ Hamstrings
- ✓ Calf
- ✓ Gluteals

Pain Characterization

This step involves a detailed analysis of your pain. You will use a sliding scale from 0 to 10 to indicate your pain level, which is accompanied by descriptive labels and emojis to help you accurately communicate your experience, ranging from minimal discomfort to extreme pain. You will then describe the quality of your pain by selecting descriptors like sharp, dull, or burning. Finally, you will provide

context on the pain's duration and how frequently it occurs, painting a complete picture of your symptoms.

Use this scale consistently for tracking:

Scale	Level	Description
1	Low	Barely noticeable, doesn't affect function
2	Low	Barely noticeable, doesn't affect function
3	Low	Barely noticeable, doesn't affect function
4	Moderate	Noticeable but tolerable, minor interference
5	Moderate	Noticeable but tolerable, minor interference
6	Moderate	Noticeable but tolerable, minor interference
7	Severe	Significantly limits activities, very distressing
8	Severe	Significantly limits activities, very distressing
9	Severe	Unbearable, unable to function
10	Severe	Unbearable, unable to function

Range of Motion Assessment

To objectively measure your functional limitations, the app will guide you through a range of motion assessment. Using your device's camera and pose estimation technology, it will analyze your movement to determine specific angle limitations. This data helps the system generate more accurate and personalized recommendations that respect your current physical capabilities.

Muscle History

Providing historical context ensures your plan is safe and informed. In this section, you will document any previous injuries to the same area, past treatments or therapies you have undergone, and any current medications you are taking. This information helps the app avoid recommending activities that could aggravate old injuries and ensures your new plan builds appropriately on your past experiences.

Summary Review

Before generating your plan, you are presented with a comprehensive summary of all the information you have entered. You can review every detail, from your selected goal to your pain history. If you need to make any changes, you can tap the back button to navigate to any previous section and edit your responses. Once you have verified that everything is correct, you tap the "Generate Plan" button to proceed.

Plan Generation

Upon your confirmation, the app automatically processes all your assessment data through its intelligent algorithms. It first analyzes your inputs to understand the nuances of your condition. It then generates a structured 4-week exercise plan, pulling from a comprehensive database of proven rehabilitation exercises. Simultaneously, it creates a set of treatment recommendations—such as ice, heat, or rest—based on your reported pain severity and other factors. Your entire plan, including all exercises and treatments, is immediately saved to

your private profile within the app, making it instantly accessible from your dashboard.

Special Cases: Your safety is the top priority. If you report a combination of severe pain and a recent injury, the app will follow clinical best practices by skipping the generation of a strenuous exercise plan. Instead, it will provide a focused set of treatment-only recommendations aimed at managing the acute phase of your injury. When this occurs, a clear warning dialog will appear, explaining the reasoning behind this conservative approach to ensure you understand the importance of rest and specific treatments before beginning more active exercises.

3.4 Daily Pain Tracking

Track your pain levels daily to monitor progress:

1. Navigate to Daily Assessment (from dashboard notification or menu).
2. ROM: Range of Motion.
   - ✓ App may prompt you to record range of motion
   - ✓ Use front camera with pose detection
   - ✓ Follow on-screen instructions for movement
3. Optional: Use the pain slider.
   - ✓ Drag to select current pain level (0-10)
   - ✓ Emoji and description update in real-time
4. Tap "Complete" to save.

Best Practices:

- ✓ Track pain at the same time each day
- ✓ Be consistent with your rating scale
- ✓ Note any significant changes in the notes field

3.5 Exercise & Treatment Management

This section of the app is your central hub for viewing and personalizing your rehabilitation plan. You can see all your assigned exercises and treatments, make adjustments as you progress, and browse the full library of available exercises.

Viewing Your Plan

From the "Plan" tab, you can view your complete rehabilitation plan. All your assigned exercises are clearly organized by week, allowing you to see your progression at a glance. Alongside your exercises, you can review your treatment recommendations, each with a detailed description of its purpose and application. To help personalize your routine, you can add personal notes to any exercise, such as reminders about form or how the exercise felt on a particular day.

Replacing Exercises

The app provides distinct methods for modifying your exercise plan. If you need to swap out an assigned exercise, you can use the Replace Exercise feature. When you select this option for a specific exercise card, the system automatically removes the current exercise and replaces it with a new one that is still appropriate for your condition, chosen from the same filtered set of exercises used during your initial assessment. This change ensures your plan remains consistent with your rehabilitation goals.

Adding Exercises

To add a completely new exercise to your plan, you use the Add Exercise function. This feature presents you with two distinct options. The first option, "Choose from Library," allows you to browse the full exercise library to manually select and add an exercise. The second option, "Add Custom Exercise," opens a form for you to create a completely new exercise from scratch. You will fill in details such as the exercise name, description, and parameters like sets, reps, or

hold duration. Upon saving, this custom exercise is added directly to your personal library and inserted into your current plan.

Managing Treatments

To ensure your treatment recommendations stay aligned with your current condition, you can tap the "Refresh Treatments" button. This action regenerates your treatment plan based on your most recent assessment data, which is particularly useful after you have made significant progress or experienced a change in your pain level. Furthermore, you can add optional treatments to your plan. By tapping "Add Optional Treatment," you can browse a list of supplementary recommendations and select those you wish to incorporate, allowing you to tailor your recovery process beyond the core generated plan.

3.6 Recording Exercise Sessions

Recording sessions combine video capture with AI-powered pain detection to help you safely monitor your exercises and track your progress. This feature uses your device's camera to provide real-time feedback during your rehabilitation workouts.

Pre-Recording Setup

To begin recording, you first navigate to the Record tab from the main navigation bar. The app will then present you with important instructions to ensure optimal recording conditions, including ensuring adequate lighting in your space, clearing sufficient area around you for movement, and positioning your camera to capture your full body for accurate pose detection. If this is your first time using the recording feature, the app will request camera permission, which you must grant to proceed. Before starting your main exercise, you have the option to complete a guided warmup by tapping the "Start Warmup" button, which will lead you through a structured stretching routine with clear timer and instructions, though you can skip this step if you are already properly warmed up.

During Recording

Once you begin recording, the main screen presents several key interface elements. The Heads-Up Display at the top of the screen shows the current exercise name along with your current set and repetition count, a running stopwatch timer tracking your session duration, and a visual progress indicator. The central portion of the screen displays the live camera preview from your device, with buttons available to toggle between front and back cameras and activate the flashlight if your device supports it.

After Recording

Upon completing your session, the Confirm & Save screen allows you to review important session metrics including total duration, average pain level, peak pain detected, and sets/reps completed. You can add optional session notes to document how the exercise felt or any observations about your form. The screen also includes an option to rate your perceived exertion before saving the session to your exercise history. If you're not satisfied with the recording, you can choose to discard it instead. Finally, the app offers an optional guided cooldown routine with stretching exercises and timers similar to the warmup, helping prevent soreness and promote recovery after your workout.

3.7 Reports & Analytics

The Reports section provides comprehensive insights into your rehabilitation progress, allowing you to track your journey and understand patterns in your recovery. This feature consolidates data from your assessments, exercise sessions, and daily pain tracking to present meaningful analytics about your improvement over time.

Plan Overview

The Rehabilitation Plan Overview presents a summary of your current rehabilitation strategy, including a detailed week-by-week breakdown that shows your planned progression. This report displays your exercise completion status to help you stay on track with your routine and monitors your treatment adherence to ensure you're following all recommended therapies alongside your exercises.

Exercise Calendar

An Exercise Calendar provides a grid view of all your recorded sessions, organized by date with color-coding that immediately shows your completion status for each day. You can tap on any day within the calendar to access a detailed breakdown of what exercises you performed, how long you exercised, and any pain levels recorded during those sessions.

Pain Level Chart

The Pain Level Chart displays a line graph that visualizes your pain trends over time, allowing you to identify patterns in your discomfort levels. You can adjust the view to see daily, weekly, or monthly perspectives, and the chart can overlay your exercise sessions to help you understand the correlation between your activity levels and pain experiences.

Progress Statistics

Progress Statistics consolidate key metrics including your total exercises completed, average pain reduction percentage, current active days streak, and any range of motion improvements if you've been tracking this metric through the pose detection features.

Exporting Reports

To generate a PDF report, simply tap the "Export PDF" button and wait for the system to aggregate your data, during which a progress indicator will show the preparation status. Once complete, you can review the generated PDF preview before finalizing. On mobile devices, you can share the report through your system's share dialog to email, messaging apps, or cloud storage services, while on web platforms you can download or print the document directly.

The PDF report contains comprehensive information including a header with your personal information, a data freshness timestamp indicating when the information was last updated, an assessment summary table, your key progress statistics, pain trend charts, a detailed exercise history table, your current treatment recommendations, and an optional custom notes section for additional observations.

3.8 Profile & Settings

The Profile & Settings section serves as your central hub for managing your account, personalizing your experience, and controlling your data within the application. This area allows you to review your personal information, adjust notification preferences, and perform important account management tasks.

Profile Information

The profile screen prominently displays your basic account details, including your display name and registered email address. You can tap on your profile photo to change or update your picture, which helps personalize your account. The interface also shows motivational statistics such as your current days active streak, encouraging consistent engagement with your rehabilitation program. Additionally, you can review metadata about your current rehabilitation plan, giving you quick access to information about your active treatment schedule and exercise regimen.

Settings Options

Within the Reminders section, you can customize various notification settings to help you stay on track with your rehabilitation. You can enable or disable daily assessment notifications that prompt you to log your pain levels, set exercise session reminders to ensure you complete your scheduled workouts, and receive treatment adherence alerts for your prescribed therapies. The system also allows you to set custom reminder times that fit your daily schedule and preferences.

The Data Management section provides tools for controlling your information. The Export All Data feature lets you download a complete backup of your assessment history, exercise records, and treatment plans for your personal records or to share with healthcare providers. If you need to free up storage space on your device, the Clear Cache option removes temporary files while preserving your essential data. For users who want to revisit the app's guidance, the Reset Tutorials feature allows you to replay all onboarding flows and educational content.

Under Account Actions, you can make important changes to your account configuration. The Edit Profile function allows you to update your name, email address, and profile photo as needed. For security purposes, the Change Password option enables you to update your login credentials. If you started using the app as a guest, the Manage Guest Mode feature helps you convert your guest session to a full registered account to enable cloud synchronization and data backup. The Log Out option safely signs you out of the application and clears local data from your device while preserving your account on the server.

3.9 Data Synchronization

PocketPT uses an intelligent offline-first architecture that ensures your data remains accessible even without an internet connection while automatically synchronizing when connectivity is available. This approach provides a seamless experience whether you're online or offline, with your information securely stored and synchronized across devices.

How Sync Works

The application employs Automatic Sync to keep your data current across all platforms. Any changes you make are immediately saved to your device's local storage using Hive database technology, ensuring no data loss even if you lose connectivity unexpectedly. Background synchronization with Firebase occurs automatically in several scenarios: when you return the app from the background on mobile devices, approximately every five minutes while the app remains active, and immediately after you complete major actions such as finishing an assessment or completing an exercise recording session.

For situations where you need immediate synchronization, Manual Sync options are available. You can perform a pull-to-refresh gesture on the Dashboard screen to force an immediate synchronization, or you can navigate to Profile > Diagnostics and select the "Force Sync Now" option for a comprehensive data sync that ensures all your latest information is backed up to the cloud.

The application maintains full Offline Operation capabilities for most features, allowing you to continue using the app without interruption when internet connectivity is unavailable. All core functionality works offline except for a few specific features: initial login and registration processes require internet connectivity, PDF export functionality needs to fetch the latest data from cloud services, and custom exercise image uploads depend on cloud storage access.

The system employs intelligent Conflict Resolution to handle situations where the same data has been modified in multiple locations. A last-write-wins strategy automatically resolves most conflicts by prioritizing the most recent changes, while the application will show manual intervention options in the Diagnostics section if significant conflicts are detected that require your attention to resolve.

INSTALLATION GUIDE

4.1 Android Installation (APK)

For Users

Prerequisites:

- ✓ Android device running Android 7.0 (Nougat) or higher
- ✓ Google Drive app installed (for download link access)

Installation Steps:

1. **Receive Installation Link:** You will receive a Google Drive link from your administrator or therapist (e.g., https://drive.google.com/file/d/.../view?usp=sharing).

2. **Download APK:** Open the link on your Android device, tap the download icon (down arrow), and wait for the PocketPT-v<version>.apk file to download.

3. **Enable Unknown Sources:** Navigate to your device's Settings > Security and enable "Install unknown apps" for your browser or the Files app. Alternatively, tap the downloaded APK file and follow the on-screen permission prompt.

4. **Install Application:** Open your Downloads folder, tap the PocketPT-v<version>.apk file, and then tap "Install." Wait for the installation to complete.

5. **Grant Permissions:** Open the PocketPT app. When prompted, grant Camera permission for recording and pose detection. It is also recommended to grant Notifications permission for reminders.

6. **First Launch:** The app will initialize its data systems (Firebase and Hive storage). The login screen will then appear, where you can create a new account or sign into an existing one.

Updating the App

To update, simply download the newer APK version from the updated Google Drive link provided by your administrator. You can install it over the existing version without uninstalling first (if it uses the same signature). Your existing data will be preserved.

For Administrators

Building the APK:

1. Clean previous builds: $flutter\ clean$
2. Get dependencies: $flutter\ pub\ get$
3. Build release APK: $flutter\ build\ apk\ --release$

Optional: Use the provided PowerShell script (.\copy_apk.ps1) to move the generated APK from build\app\outputs\flutter-apk\ to a more convenient location.

Distributing via Google Drive:

1. Create a shared folder named PocketPT Releases.
2. Upload the app-release.apk file.
3. Set the sharing permissions for the file to "Anyone with the link" (Viewer).
4. Copy the shareable link and distribute it to users via email or messaging.

4.2 Web Installation (GitHub Pages)

For Users

1. Open your web browser.
2. Navigate to: https://cosmicmashups.github.io/PocketPT/

Bookmark the page for easy access. No installation is required as the app runs directly in your browser.

Adding to Home Screen (Mobile Browsers):

Chrome Android:

1. Open the PocketPT web app.
2. Tap the menu (three dots).
3. Select "Add to Home screen."
4. Name the shortcut and tap "Add."

Safari iOS:

1. Open the PocketPT web app.
2. Tap the share icon.
3. Select "Add to Home Screen."
4. Name the shortcut and tap "Add."

For Developers

PocketPT uses GitHub Actions for continuous deployment.

1. Ensure GitHub Pages is enabled in your repository settings (Settings > Pages), set to deploy from the gh-pages branch from the root folder.

2. The workflows (.github/workflows/deploy-web.yml and flutter-web-deploy.yml) will trigger automatically on every push to the main branch.

3. Monitor the deployment progress and check for any errors in the logs under the "Actions" tab in GitHub.

Manual Deployment:

Use the provided PowerShell script for a streamlined process: $.\backslash deploy_web.ps1$

This script automates the following steps:

1. Cleans previous builds.
2. Runs flutter pub get to fetch dependencies.
3. Builds the web app with the correct base href (--base-href /PocketPT/).
4. Adds a .nojekyll file to prevent Jekyll processing.
5. Creates a 404.html file to support Single Page Application (SPA) routing.
6. Force-adds the build/web/ directory to Git, creates a temporary commit, splits the subtree to the gh-pages branch, and force-pushes it to the remote repository.
7. Cleans up the temporary branch and commits.

Manual Command Sequence:

If you prefer to run the commands manually, use this sequence:

```
# Clean and prepare

flutter clean

flutter pub get

# Build for web

flutter build web --release --base-href /PocketPT/

# Add required files for GitHub Pages

echo > build/web/.nojekyll

cp build/web/index.html build/web/404.html

# Deploy to gh-pages branch

git add -f build/web/
```

```
git commit -m "chore(deploy): web build"

git subtree split --prefix build/web -b gh-pages-temp

git push origin gh-pages-temp:gh-pages --force

# Cleanup

git branch -D gh-pages-temp

git reset --soft HEAD~1

git restore --staged build/web
```

Verification:

After deployment, wait 2-5 minutes for GitHub Pages to update. Then, visit https://cosmicmashups.github.io/PocketPT/ and perform these checks:

1. Check the browser console for any JavaScript errors.
2. Verify all assets load correctly using the browser's Network tab.
3. Test core functionality, including login/registration, the assessment flow, recording (ensure camera permission works), and reports generation.

Troubleshooting Deployment:

Issue	Solution
404 errors when navigating to deep links	Verify 404.html exists in the build output and its content matches index.html.
Assets (CSV, models) not loading	Confirm the --base-href parameter exactly matches your repository name (case-sensitive).
Blank white screen on load	Clear the browser cache and check the console for specific error messages.
GitHub Actions workflow fails	Review the action logs and ensure the Flutter version specified in the workflow matches your local development environment.

4.3 Local Development Setup

Prerequisites

Before beginning development, ensure you have the following required software installed on your system:

- ✓ Flutter SDK on the stable channel, compatible with Dart version ^3.7.2
- ✓ Android Studio for Android builds, emulators, and platform tooling
- ✓ Xcode (macOS only) for iOS builds and simulator access
- ✓ Visual Studio Code or Android Studio IDE with Flutter/Dart extensions
- ✓ Git for version control and repository management

Firebase Setup:

To configure Firebase for your development environment, follow these steps:

1. Create a new Firebase project at console.firebase.google.com
2. Add an Android app with the package name com.example.pocketpt
3. Download the generated google-services.json file and place it in the android/app/ directory
4. Add a web app to your Firebase project

5. Generate the required firebase_options.dart file using the FlutterFire CLI by running:

```
flutterfire configure
```

Setup Steps

1. **Clone Repository:** Begin by cloning the PocketPT repository to your local machine using:

```
git clone
https://cosmicmashups.github.io/PocketPT.git
cd PocketPT
```

2. **Install Dependencies:** Fetch all required Flutter packages and dependencies by executing:

```
flutter pub get
```

3. **Run Code Generation:** If any Hive data models have been modified, regenerate the necessary code with:

```
flutter pub run build_runner build --delete-conflicting-outputs
```

4. **Verify Setup:** Check that your development environment is properly configured by running:

```
flutter doctor
```

Address any issues or warnings flagged by the command to ensure all components are correctly installed.

5. **Run App:** Launch the application using one of the following methods depending on your target platform.

Platform	Code
Mobile Emulator	flutter run
Specific Device	flutter devices flutter run -d <device-id>
Web Development	flutter run -d chrome --web-port 6006 --web-renderer canvaskit

Development Workflow

Hot Reload: During active development with flutter run running, you can quickly apply changes by:

- ✓ Pressing r in the terminal where Flutter is running
- ✓ Or saving files in your IDE if hot reload is enabled, which automatically refreshes the running application

Testing: Execute various test scenarios to ensure code quality and functionality:

```
# Run all tests in the project
flutter test

# Run a specific test file
flutter test test/unit/exercise_service_test.dart

# Run tests with coverage reporting
flutter test --coverage
```

Building: Create production-ready builds for different platforms using these commands:

```
# Android APK for direct installation
flutter build apk --debug

# Android App Bundle for Google Play Store distribution
flutter build appbundle --debug

# Web deployment build
flutter build web --debug --base-href /PocketPT/
```

Additional Development Notes:

- ✓ The app uses Riverpod for state management, so familiarize yourself with provider patterns
- ✓ Hive is used for local storage with the main box named rehabBox
- ✓ ML features require proper asset configuration (pain_recognition_model.ptl in assets)
- ✓ For web development, ensure proper base href configuration matches your deployment path
- ✓ Guest mode functionality provides offline capability when Firebase is unavailable

TROUBLESHOOTING

5.1 Frequently Asked Questions

General

Q: Is PocketPT free?

A: The core app is free. Premium features may require subscription.

Q: Can I use PocketPT without internet?

A: Yes! The app is offline-first. You need internet only for initial login and cloud sync.

Q: Is PocketPT a replacement for physical therapy?

A: No. PocketPT is a supplement to professional care, not a replacement. Always consult healthcare providers.

Recording

Q: Do I need to record every exercise?

A: Not required, but highly recommended. Pain detection and progress tracking work best with recordings.

Q: Are my recordings uploaded to the cloud?

A: No. Videos are processed locally and immediately discarded. Only session metrics are saved.

Data & Privacy

Q: Who can see my data?

A: Only you. Healthcare providers can only see data you explicitly share (e.g., exported PDF).

Q: What happens to my data if I delete the app?

A: Local data is removed. Cloud data persists until you delete your account.

Q: Can I export my data?

A: Yes. Profile > Data Management > Export All Data (JSON format for full backup).

Technical

Q: Why is my app slow?

A: Try clearing cache (Profile > Data Management > Clear Cache) and closing background apps.

Q: How much storage does PocketPT use?

A: Typically 50-150MB depending on exercise history. Regular cache clearing helps.

Q: What if I get a new device?

A: Log in with the same account on your new device. Data will sync automatically.

5.2 Common Installation Issues

Android: "App not installed"

This error typically occurs when there's a conflict with an existing installation, insufficient storage space, or a corrupted APK file download. The package name "com.example.pocketpt" might already exist on your device from a previous installation, or your device may not have enough free space to complete the installation process.

To resolve this issue, follow these steps:

1. First, uninstall any existing PocketPT version by running this command if you have ADB access: adb uninstall com.example.pocketpt
2. Ensure you have at least 750MB of free storage space available on your device
3. Re-download the APK file and verify it's the correct size (typically 500-750MB)
4. As an alternative installation method, try using ADB directly: adb install PocketPT-v1.0.0.apk

Web: Blank Screen or Loading Forever

When the web version displays a blank screen or loads indefinitely, this is usually caused by browser caching issues, incorrect URL casing, or missing assets due to deployment problems.

Follow this troubleshooting sequence:

1. Clear your browser cache completely by pressing Ctrl+Shift+Delete and selecting "Cached images and files"
2. Open browser developer tools (F12) and check the console for JavaScript errors, particularly CORS issues or missing asset files
3. Verify the URL exactly matches the case used during build deployment (e.g., /PocketPT/ not /pocketpt/)
4. Temporarily disable browser extensions, especially ad blockers that might interfere with asset loading
5. Test in incognito or private browsing mode to rule out extension conflicts

5.3 Authentication Issues

"Email verification failed"

Email verification issues commonly stem from email delivery delays, incorrect email addresses, or provider-specific filtering problems.

To address verification failures:

1. Check your spam or junk folder thoroughly for the verification email
2. Wait 2-3 minutes then use the "Resend" function (maximum of 3 attempts allowed)
3. Double-check that you entered your email address correctly during registration
4. Try using an alternative email provider if delays persist with your current provider

"Firebase authentication error"

Authentication errors typically relate to connectivity issues, Firebase project configuration, or disabled sign-in methods.

Resolution steps include:

1. Verify you have a stable internet connection
2. Ensure the Firebase project is active and properly configured (for developers)

3. Confirm that email/password sign-in is enabled in Firebase Console under Authentication > Sign-in method
4. Use guest mode temporarily as a workaround (web version only)

"Automatically signed out"

Automatic sign-outs are normal security measures that occur after approximately 1 hour of inactivity, or if your account was modified elsewhere.

Simply log in again using your credentials. If you've forgotten your password, use the password reset function to regain access to your account.

5.4 Camera & Recording Issues

"Camera permission denied"

Camera functionality is essential for recording exercises and pose detection, but requires explicit user permission.

On Android devices:

1. Navigate to Settings > Apps > PocketPT > Permissions
2. Enable Camera access
3. Fully restart the application

For web browsers:

1. Click the lock or information icon in your browser's address bar
2. Set Camera permissions to "Allow"
3. Refresh the web page to apply the new permissions

"Pain detection not working"

The facial pain recognition feature requires specific conditions to function properly, including adequate lighting and proper camera orientation.

Symptoms include no colored overlay during recording or persistent "Processing..." messages.

Troubleshooting steps:

1. Switch to the front-facing camera since pain detection only works with frontal facial views
2. Ensure good lighting conditions so the model can clearly detect facial features
3. For developers, verify that assets/model/pain_recognition_model.ptl exists in the build and rebuild if necessary
4. Confirm camera permissions are granted (see previous section)
5. Try on a different device if possible, as older phones may lack PyTorch Mobile support

"Recording freezes or lags"

Performance issues during recording can result from insufficient device resources or camera configuration problems.

To improve recording performance:

1. Close other running applications to free up RAM and processing power
2. Lower the camera resolution through developer settings if available
3. Temporarily disable pain detection if performance remains poor
4. Restart your device to clear temporary memory issues

5.5 Data Sync Issues

"Data not syncing across devices"

The offline-first architecture syncs data periodically, but requires internet connectivity and proper authentication.

To diagnose sync problems:

1. Navigate to Profile > Diagnostics to check the Last Sync timestamp
2. Use the "Force Sync Now" function to trigger an immediate synchronization
3. Monitor for any error messages that appear during the sync process

Resolution steps include:

1. Ensure you have an active internet connection, trying mobile data if WiFi is unstable
2. Verify you're logged into the same Firebase account on all devices
3. Wait for automatic sync cycles (occur approximately every 5 minutes)
4. Use pull-to-refresh on the Dashboard to trigger a manual sync
5. For developers, check Firebase quota limits to ensure Firestore hasn't exceeded free tier limitations

"Data disappeared after update"

While rare, data loss can occur due to Hive box corruption or incomplete synchronization before app updates.

Data recovery options:

1. Log in with your original account and wait for automatic data restoration from Firebase
2. Check if you have backups available through Profile > Data Management > Restore from Backup
3. "Changes lost when switching devices"
4. Prevent data loss by following these practices:
5. Always wait for the "Data synced" confirmation message before closing the application
6. Enable Background Sync in your device settings (mobile versions)
7. Avoid force-closing the app immediately after making important changes

5.6 Performance Issues

"App is slow or laggy"

Performance degradation can result from cache buildup, insufficient device resources, or animation overload.

Performance improvement strategies:

1. Clear the app cache through Profile > Data Management > Clear Cache
2. Restart your device to free up system resources
3. Update to the latest app version for performance optimizations
4. Ensure your device meets minimum requirements (2GB RAM, Android 7.0+)
5. For developers, adjust kDefaultAnimationDuration in theme settings to reduce animation intensity

"Excessive battery drain"

Battery consumption issues typically stem from camera usage, continuous pain detection, or frequent synchronization attempts.

To conserve battery life:

1. Ensure you properly close recording screens when not actively using them
2. Disable background sync if you don't require real-time data synchronization
3. Reduce notification frequency in app settings
4. Update to the latest app version, which includes ongoing battery optimization improvements

5.7 Report Generation Issues

"PDF export failed"

PDF generation relies on data availability, internet connectivity, and proper service configuration.

When export fails, tap the error message to see specific details. Common scenarios include:

- "No internet connection": Connect to WiFi or mobile data and retry
- "Failed to fetch latest data": Wait a few minutes as servers may be temporarily busy, then attempt again
- "Insufficient data": Complete at least one full assessment and recording session before generating reports
- "PDF generation error": Clear your app cache and retry; report persistent issues as bugs

"PDF is empty or incomplete"

Incomplete reports typically indicate missing foundational data required for comprehensive reporting.

Ensure you have completed these prerequisites:

1. At least one full assessment with plan generation
2. Multiple recorded exercise sessions with proper completion
3. Several daily pain tracking entries for trend analysis
4. Force sync before exporting by using Profile > Diagnostics > Force Sync
5. Wait approximately 30 seconds after synchronization completes before attempting export

5.8 Error Codes Reference

This reference table helps identify specific error conditions and appropriate responses:

Code	Meaning	Recommended Action
AUTH_001	Invalid credentials	Verify your email and password are correct; use password reset if necessary
AUTH_002	Email not verified	Check your email inbox and spam folder; resend verification if needed
SYNC_001	Network timeout	Check your internet connection stability and retry after 1 minute
SYNC_002	Firebase permission denied	Contact administrator; your account may have been temporarily disabled
DATA_001	Hive box corruption	The application will attempt automatic repair; restart if the issue persists
DATA_002	Data integrity check failed	Force a synchronization to recover data from cloud backups

CAMERA_001	Camera initialization failed	Restart the application and ensure camera permissions are granted
MODEL_001	Pain recognition model load failed	Reinstall the application; the model file may be missing or corrupted
PDF_001	Export service unavailable	Temporary server issue; wait a few minutes and try the export function again

APPENDIX

6.1 Glossary

- **AROM.** Active Range of Motion—movement produced by your own muscle contraction.
- **Assessment.** Initial evaluation to determine rehabilitation needs.
- **CSV Database.** Comma-separated values file containing exercise library.
- **Firebase.** Cloud backend for authentication and data sync.
- **Guest Mode.** Temporary session without cloud account.
- **Hive.** Local database used for offline-first storage.
- **Pain Detection.** AI-powered facial analysis during exercise recording.
- **Pose Detection.** ML-based body landmark tracking for form analysis.
- **Rehabilitation Plan.** Personalized 4-week exercise program.
- **ROM.** Range of Motion—how far a joint can move in each direction.
- **Sync.** Process of uploading local data to cloud and downloading updates.
- **Treatment Plan.** Recommended therapies and interventions based on severity.

6.2 Privacy & Data

What data is collected:

- ✓ Account information (email, name)
- ✓ Assessment responses
- ✓ Exercise performance metrics
- ✓ Pain ratings
- ✓ Camera recordings (processed locally, not stored)

Data storage locations:

- ✓ Local: Hive database on device
- ✓ Cloud: Firebase Firestore (encrypted in transit and at rest)

Data retention:

- ✓ Account data: Retained until account deletion.
- ✓ Exercise history: Retained for progress tracking.
- ✓ Recordings: Metadata only; videos not uploaded.

6.3 Support & Feedback

For support and to provide feedback on your experience with PocketPT, please feel free to reach out via email. You can contact our support team directly at bym0866@dlsud.edu.ph for assistance with any technical issues, questions about app functionality, or to share suggestions for improvement. We value user feedback and are committed to continuously enhancing the application based on your input.

Reporting Bugs:

1. Note the error code (if shown)

2. Screenshot the issue

3. Describe steps to reproduce

4. Submit via e-mail: bym0866@dlsud.edu.ph.

6.4 Credits

PocketPT Development Team:

- ✓ Yuri M. Brown
- ✓ Joseph II S. Lina
- ✓ Francis Vincent J. Panganiban

Third-Party Libraries:

- ✓ Flutter (Google)
- ✓ Firebase (Google)
- ✓ Hive (Isar)
- ✓ PyTorch Mobile (Meta)
- ✓ pdf package (DavBfr)
- ✓ Riverpod (Remi Rousselet)

Assets:

- ✓ Icons: Material Design Icons (Apache 2.0)

CONCLUSION

This manual covers the complete PocketPT experience from installation through advanced feature usage. For the latest updates, troubleshooting tips, and video tutorials, visit the support website or check the in-app Help section.

Remember: PocketPT is a supplement to professional physical therapy, not a replacement. Always consult with a licensed healthcare provider before starting any rehabilitation program.